Name: Student ID:

Read the following scenarios and answer the questions about hypothesis testing errors.

1. A large pharmaceutical company has a new drug to relieve the symptoms of rheumatoid arthritis, a condition that causes joint stiffness and pain, but it does have side-effects.

$H_0$: The new drug is no different than a placebo in relieving symptoms.

$H_A$: The new drug is different than a placebo in relieving symptoms.

- Describe a Type I error in this case.

- Describe a Type II error in this case.

- Which error do you think has greater real-world consequences? Argue in two sentences.

2. Municipal water authorities routinely sample the water supply to determine the level of lead in the water. Elevated lead levels can lead to severe developmental problems in children.

$H_0$: The lead levels are at an acceptable level.

$H_A$: The lead levels exceed acceptable levels.

- Describe a Type I error in this case.

- Describe a Type II error in this case.

- Which error do you think has greater real-world consequences? Argue in two sentences.

3. A food safety inspector is called upon to investigate a restaurant with a few customer reports of poor sanitation practices. The food safety inspector uses a hypothesis testing framework to evaluate whether regulations are not being met. If he decides the restaurant is in gross violation, its license to serve food will be revoked.

- Write the null hypothesis in words.

- Write the alternative hypothesis in words.

- Describe a Type I Error in this case.

- Describe a Type II Error in this case.

- Which error is more problematic for the restaurant owner? Why?

- Which error is more problematic for the diners? Why?

- As a diner, choose a significance level ($\alpha$) that you prefer that the food safety inspector uses in their hypothesis test to establish evidence of health concerns (and then revoke a restaurant's license). Explain your reasoning in two to three sentences.